

PRACTICAL COURSE WORKBOOK

Biomedical Data Analysis with Python

Work with digital-health information responsibly

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	BioPython, Genomics, Pandas, Clinical data, R, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use BioPython appropriately in a realistic, bounded task.
- Use Genomics appropriately in a realistic, bounded task.
- Use Pandas appropriately in a realistic, bounded task.
- Use Clinical data appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Biomedical Data Analysis with Python is a structured, practical course covering BioPython, Genomics, Pandas, Clinical data, R. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 BioPython: Health context	1. Care pathways with BioPython 2. Health data with Genomics 3. Stakeholders with Pandas
02 Genomics: Digital systems	1. Standards with Clinical data 2. Interoperability with R 3. Workflows with BioPython
03 Pandas: Quality and safety	1. Validation with Genomics 2. Privacy with Pandas 3. Human oversight with Clinical data
04 Clinical data: Applied evaluation	1. Implementation with R 2. Monitoring with BioPython 3. Equity with Genomics

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Biomedical Data Analysis with Python project using BioPython, Genomics, Pandas.

DELIVERABLE	A working Biomedical Data Analysis with Python artefact plus an evidence-based self-review.
TOOLS	A suitable BioPython environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using BioPython.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21
<https://www.who.int/publications/item/9789240116870>

FHIR specification

HL7 International - reviewed 2026-08-21
<https://hl7.org/fhir/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the BioPython scope.